

BIOL-8 Comprehensive Final Exam

College of the Redwoods, Del Norte

Instructor: Dr. Daniel Friedman

Name: _____

Date: _____

Modules 01–17 (cumulative)

Total Points: 100

Answer **all** questions in **Parts A and B**. In **Part C**, answer **exactly five** of the **seven** prompts (each is worth **5** points). For **Part D**, choose **one** essay prompt.

Part A: Multiple Choice (51 points)

Each question is worth 1 point.

Module 01 — Exploring Life Science

1. A scientific hypothesis earns its usefulness because it is structured to be:

- A) Locked after one experiment
- B) Shared only among experts privately
- C) Addressed with observations or experiments that may support or refute predictions
- D) Equivalent to rumor until celebrities endorse it

2. Typical levels inside one multicellular vertebrate ascend from grouped cells upward as:

- A) Lysosome toward ecosystem
- B) Allele toward gamete
- C) Cell toward tissue toward organ toward organ system
- D) Chloroplast toward virus

3. Natural selection reshapes allele frequencies largely through:

- A) Osmosis replacing meiosis
 - B) Lysosomes stripping chromosomes nightly
 - C) Chlorophyll in human erythrocytes
 - D) Uneven survival and reproduction linked to heritable variants
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Module 02 — Chemistry of Life

4. Atomic **mass number** for a stated isotope counts:

- A) Ionic charge squared
- B) Electron shells minus neutrons
- C) Nuclear protons **plus** nuclear neutrons in that nucleus
- D) Molecular weight of seawater only

5. Table salt (NaCl) crystals illustrate attraction between opposite ions mainly through:

- A) Ionic bonding between Na^+ and Cl^- ions in a lattice
- B) Metallic delocalization only
- C) Neon dispersion only
- D) Oxygen double bonding like O_2 only

6. Compared with distilled water near pH 7, a solution registering **pH 11** in intro framing behaves as:

- A) Neutral
- B) Strongly acidic
- C) Mildly acidic

- D) Basic / alkaline (pH above 7)
-

Module 03 — Biomolecules

7. Digestive enzymes breaking polymers by adding water exemplify:

- A) Osmosis
- B) Passive diffusion only
- C) Competitive inhibition universally
- D) Hydrolysis

8. **Steroid** signaling hormones are grouped chemically foremost as:

- A) Nucleic acid archives
- B) Cellulose structural fibers only
- C) Lysosomal tags only
- D) Lipids (steroids)

9. Protein **primary** structure names:

- A) Quaternary packing of several chains only
 - B) Lysosomal proton gradients only
 - C) The ordered sequence of amino acids along a polypeptide
 - D) Surface glycolipids only
-

Module 04 — Cellular Function

10. Rough endoplasmic reticulum appears studded with cytosolic:

- A) Lysosomes digesting the nucleus
- B) Ribosomes co-translating secretory and membrane proteins
- C) Chloroplast DNA replication in all animal cells
- D) Mitochondrial ribosomes attaching to Golgi

11. Lysosomes mainly:

- A) Produce bulk aerobic ATP
- B) Carry photosynthetic pigments
- C) Fuse only with mitochondria
- D) House acidic hydrolytic enzymes degrading engulfed material and aging organelles

12. Compared with spinach mesophyll cells, typical mammal skeletal muscle fibers lack prominent:

- A) Genetic material
 - B) Chloroplast-containing photosynthesis organelles
 - C) Plasma membranes
 - D) Translating ribosomes
-

Module 05 — Membranes

13. The **fluid mosaic** model summarizes plasma membranes as:

- A) Mosaic virus coats only
- B) Calcium phosphate trabecula only
- C) Lipid bilayer with drifting proteins plus diverse surface carbohydrate trees
- D) Peptidoglycan bacterial walls

14. **Osmosis** refers especially to:

- A) Net diffusion of chloride ions through open membrane channels only
- B) Genetic drift sampling
- C) Oxygen bound to hemoglobin
- D) Diffusion of water toward osmotic balance across selective barriers

15. **Isotonic** bathing fluid relative to mammal red blood cells in intro cartoons minimizes:

- A) Osmosis stopping completely
- B) Large-scale cell swelling versus pure water extremes or large-scale shrinking versus very salty extremes

- C) All membrane protein motion
 - D) Gene transcription entirely
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Module 06 — Metabolism

16. ATP is useful as short-term energy currency partly because coupling $\text{ADP} + \text{P}_i \leftrightarrow \text{ATP}$ can:

- A) Permanently erase DNA nightly
- B) Link exergonic phosphate transfers to endergonic cellular tasks when recycled in metabolism
- C) Replace hemoglobin O_2 transport
- D) Shut off every enzyme

17. Much **aerobic** ATP yield localizes to mitochondria because those organelles organize:

- A) Chlorophyll photochemistry in human liver
- B) Binary fission spindles
- C) Electron transport and chemiosmotic ATP synthase coupling
- D) Rough ER translation alone

18. **Competitive** enzyme inhibitors slow reactions mainly by:

- A) Melting nuclear lamins
 - B) Binding the active site so substrate enters less often
 - C) Digesting all substrates with lysozyme
 - D) Denaturing every protein permanently at once
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Module 07 — Genetics

19. Ribosome-guided polypeptide assembly from mRNA codons is:

- A) Replication
- B) Transcription
- C) Translation

- D) Binary fission

20. RNA polymerase extends RNA chains from a DNA template during:

- A) Mitotic metaphase
- B) Transcription
- C) Okazaki fragment sealing only
- D) Krebs cycle decarboxylation only

21. A standard chemical contrast between RNA and DNA bases is that many RNA transcripts include **uracil** where complementary DNA shows **thymine**. That means RNA often pairs adenine with:

- A) Uracil
- B) Thymine on the same RNA strand
- C) Peptide bonds
- D) Only guanine exclusively

Module 08 — Cell Division

22. Mitosis followed by cytokinesis of one diploid somatic cell typically yields:

- A) Four genetically distinct haploid nuclei
- B) Two diploid daughter nuclei ordinarily matching the parent complement (barring mutation)
- C) Gametes without meiosis
- D) One giant syncytium only

23. Crossing over between homologous chromosomes mainly occurs during:

- A) Prophase I of meiosis
- B) Cytokinesis of mitosis only
- C) Binary fission in archaea only
- D) Translation elongation

24. Human sperm and egg are ordinarily **haploid (n)** relative to **diploid (2n)** somatic cells because:

- A) Mitosis doubles chromosome number in gonads
 - B) Meiosis partitions homologs into gametes before fertilization restores zygote ploidy
 - C) Lysosomes strip one copy nightly
 - D) Viruses remove chromosomes
-

Module 09 — Tissues and Organ-System Survey

25. **Negative feedback** loops often stabilize physiological set points by:

- A) Accelerating every deviation without sensors
- B) Sensing departures from a target and activating responses that reduce the error
- C) Operating only on positive feedback systems
- D) Erasing all hormone genes

26. Chewing food plus salivary amylase beginning starch breakdown pairs:

- A) Mechanical digestion with chemical digestion
- B) Only endocrine signaling
- C) Mitosis with cytokinesis
- D) Meiosis with binary fission

27. In standard maps, **arteries** carry blood:

- A) Always toward the heart
 - B) Only inside alveoli
 - C) With no oxygen ever
 - D) Away from the heart (veins return blood toward the heart in overview diagrams)
-

Module 10 — Inheritance

28. **Alleles** are best defined as:

- A) Entire chromosomes only
- B) Alternative versions of a gene at a locus
- C) Mitochondrial ribosomes only
- D) Identical sister chromatids always

29. For one gene with complete dominance, a monohybrid **Aa** × **Aa** cross often yields offspring phenotypes near:

- A) 1 dominant : 3 recessive
- B) 9 : 3 : 3 : 1 for one gene
- C) 3 dominant-looking : 1 recessive-looking
- D) All recessive

30. Human **ABO blood type AB** illustrates **codominance** because:

- A) Only O antigens appear
 - B) Heterozygotes are always type O
 - C) Both A and B antigen alleles express together on red blood cells
 - D) Trait is Y-linked only
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Module 11 — Skeletal System

31. In adults, **hematopoiesis** mainly occurs in:

- A) Compact bone shafts only
- B) Yellow marrow exclusively for all blood lineages
- C) Red bone marrow in spongy bone regions
- D) Synovial fluid only

32. Which skeleton category includes skull bones, vertebrae, and rib cage?

- A) Appendicular

- B) Axial
- C) Synovial
- D) Intramembranous-only

33. Freely movable joints such as elbow and knee with synovial fluid lubrication classify as:

- A) Fibrous sutures cranial-only
 - B) Syndesmoses without movement
 - C) Synovial (diarthroses) permitting varied motion ranges
 - D) Physeal cartilage after epiphyseal closure
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Module 12 — Muscular System

34. **Thick** filaments in sarcomeres are composed mainly of:

- A) Actin
- B) Troponin only
- C) Myosin
- D) Titin scaffolding only

35. Sliding filament explanation of contraction holds that shortening occurs because:

- A) Thick and thin polymers melt shorter
- B) Z-lines disappear entirely
- C) Thin filaments slide past thick filaments drawing Z-discs closer
- D) New filaments polymerize during relaxation

36. The neuromuscular junction releases which neurotransmitter to initiate skeletal muscle fiber depolarization?

- A) Dopamine
 - B) Norepinephrine only universally
 - C) Glycine
 - D) Acetylcholine
-

Module 13 — Nervous System

37. The **central nervous system** consists of:

- A) All spinal nerves only
- B) Sympathetic chain ganglia only
- C) Brain and spinal cord
- D) Sensory receptors only

38. The long process carrying action potentials **away** from the soma is the:

- A) Dendrite
- B) Astrocyte foot
- C) Axon
- D) Synovial fold

39. The **sympathetic** autonomic division is associated with mobilizing the body for stress (“fight-or-flight”), including effects such as:

- A) Slowing heart rate exclusively
 - B) Promoting-only digestion exclusively
 - C) Constricting pupils exclusively
 - D) Increasing heart rate and diverting blood toward skeletal muscle in overview lessons
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Module 14 — Microbiology

40. A typical bacterial cell differs from a human cell mainly because bacterial DNA lacks:

- A) Ribosomes
- B) Plasma membrane
- C) Cytoplasm
- D) Membrane enclosing a nucleus (DNA in a nucleoid region)

41. Common bacterial growth by one cell becoming two daughters is termed:

- A) Mitosis

- B) Binary fission
- C) Meiosis
- D) Translation

42. Introductory rationales classify viruses as non-cellular partly because viruses:

- A) Lack genetic material
- B) Carry out transcription without hosts always
- C) Are invisible to microscopy
- D) Depend on host cell machinery for replication

Module 15 — Cardiopulmonary System

43. Oxygen-poor systemic venous return enters the heart first in the:

- A) Right atrium
- B) Left atrium
- C) Left ventricle
- D) Pulmonary veins

44. In alveoli, **carbon dioxide** moves predominantly:

- A) From inhaled air into arteries
- B) From pulmonary capillary blood into alveolar air to be exhaled
- C) From neurons into saliva
- D) Against partial pressure gradients always without diffusion

45. Most **oxygen** in arterial blood is carried bound to:

- A) Plasma albumin exclusively
 - B) Platelet granules
 - C) Hemoglobin
 - D) Cortisol
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Module 16 — Ecology

46. Compared with exponential (**J-shaped**) growth unchecked, logistic (**S-shaped**) growth differs mainly because:

- A) Carrying capacity is always zero
- B) Population increase slows as abundance approaches limiting resources / **K**
- C) Death rate exceeds birth rate instantly always
- D) Density-dependent factors disappear

47. Cleaner fish eating parasites off reef clients while clients gain parasite removal exemplifies:

- A) Competitive exclusion pairwise
- B) Mutualism (both organisms benefit overall)
- C) Amensalism universally
- D) Primary succession only

48. Passing energy through successive trophic levels, relative biomass and usable energy reaching top consumers tends to remain low because much energy between transfers is:

- A) Stored perfectly at apex carnivores only
- B) Increased without thermodynamic losses
- C) Converted wholly into biomass at each step
- D) Lost heavily as metabolic heat degrading achievable biomass at upper levels versus producers

Module 17 — Evolution

49. Individuals do not evolve in the population genetics sense mainly because evolution tracks:

- A) Lysosomal fusion rates
- B) Osmosis rewriting DNA
- C) Chloroplast segregation

- D) Allele-frequency change across generations

50. Dolphins (mammals) and sharks (fish) show streamlined shapes suited to marine life owing to which pattern—similar traits evolving independently rather than reflecting recent shared limb blueprints alone?

- A) Phylogeny without adaptation
- B) Genetic drift dominating ocean basins exclusively
- C) Convergent evolution producing analogous forms
- D) Sympatric speciation always

51. Homology in comparative biology commonly signals:

- A) Identical DNA sequences in every species
- B) Shared ancestry underlying similar structures even if functions diverge
- C) Traits matching only by chance with zero shared history always
- D) Convergent evolution always

Part B: Fill in the Blank (15 points)

Use the single best term from the word bank (spell as listed). Each blank is worth 1 point.

Word bank: acetylcholine, alveoli, axial, capsid, centriole, dendrite, glycolysis, hemoglobin, heterozygous, homeostasis, homology, logistic, meiosis, myosin, omnivore, osteoclast, peptidoglycan, predation, sympatric, synovial, transcription, translation

(Several terms are not used.)

1. Maintaining stable internal conditions using feedback is called _____.
2. Synthesis of RNA from a DNA template is _____.
3. Ribosome assembly of protein from an mRNA message is _____.
4. The cytosolic pathway that begins partial breakdown of glucose is _____.

5. The division process that halves chromosome number for gamete formation is _____.
6. Genotype **Aa** for one gene is _____ at that locus.
7. Skull, vertebral column, and rib cage belong to the _____ skeleton.
8. Bone-resorbing cells remodeling matrix are _____.
9. Freely movable lubricated joints such as the knee are usually _____ joints.
10. Thick filaments in sarcomeres are built mainly of the motor protein _____.
11. Motor neurons release _____ across the skeletal neuromuscular junction.
12. Branched neuron processes specialized for incoming signals are _____.
13. Gram-positive staining highlights thick _____-rich bacterial cell walls introductory comparison.
14. Many viruses possess a nucleic acid genome coated by protein called the _____.
15. Most oxygen carried in blood binds to _____ inside erythrocytes.
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Part C: Integrated Short Answer (25 points)

Choose five of the following seven questions. Write each answer in the matching Response space below (five responses total). Each response is worth 5 points. Use complete sentences.

Question numbers you are answering (list five): ____ ____ ____ ____ ____

1. Contrast **mitosis** and **meiosis** for a diploid animal cell: number of divisions, ploidy of products, crossing over involvement, genetic similarity among products, and **one** organ-level reason each process matters medically or physiologically.

2. Outline the **central dogma** (DNA → RNA → protein). Name **two** enzymes or molecular machines (**other than ribosomes**) with distinct roles along the pathway. Explain **briefly** how a nonsense **mutation** upstream can alter the working protein (**loss-of-function framing** adequate).

3. Describe **sliding-filament** contraction in skeletal muscle, explicitly mentioning **calcium** (Ca^{2+}), **ATP**, actin and myosin, and shortening that brings **Z discs** closer together. Explain briefly why paralysis of motor neurons can stop contraction even though ATP may still circulate in the blood.

4. Explain **pulmonary** circulation (movement of blood through lung capillary beds where carbon dioxide exits toward alveolar air and oxygen enters blood) versus **systemic** circulation (oxygenated blood from the left ventricle travels through arteries to tissues through capillary beds, then veins return blood toward the right atrium).

Identify where **oxygen** principally diffuses from alveolar air into pulmonary capillary blood **and** where **carbon dioxide** principally diffuses from pulmonary capillary blood toward alveolar air.

5. Explain why **antibiotic resistance** can increase in hospitals using **heritable variation**, **differences in survival** when antimicrobial drugs are used, and **differences in reproductive success** among survivors (population-level change, not an individual organism learning).

6. Define **carrying capacity (K)** in your own words for a population in a habitat with limited resources. Contrast **exponential (J-shaped)** growth with **logistic (S-shaped)** growth when a population is **far below K** versus when it is **near K**.

7. Explain why introductory biology ties **evolution** to **change in allele frequencies in a population over generations**, not to individual organisms “getting stronger.” Using **natural selection**, briefly connect **heritable variation**, **differences in survival or reproduction**, and **how trait frequencies can shift** across generations.

Response 1

Question number: _____

Response 2

Question number: _____

Response 3

Question number: _____

Response 4

Question number: _____

Response 5

Question number: _____

Part D: Essay (9 points)

Choose **one** prompt. Write multiple paragraphs (**introduction**, supporting examples, brief **conclusion**).

Option A — Ecology. Compare **exponential** and **logistic** population models; define **carrying capacity (K)**. Give **one density-dependent** population limit and **one density-independent**

limit (use examples consistent with course vocabulary). Relate **producers**, **consumers**, and **decomposers** to energy flow and explain why many trophic pyramids are wide at the base.

Option B — Evolution. Contrast **homologous** versus **analogous** traits with **one example of each**. Explain **convergent evolution** with an example from **different evolutionary lineages** facing **similar selective environments**. Connect **reproductive isolation** ideas to how populations can diverge toward separate species (introductory level is sufficient).

Option C — Integrated homeostasis. Describe how **negative feedback** plus coordinated **nervous** and **endocrine** signaling helps stabilize blood glucose **or** core temperature. Name **two organs or tissues** and state how signaling between them tightens control around a set point.

Essay option letter (A, B, or C): _____

Feedback (optional)

Notes for your instructor?

End of Comprehensive Final Exam